# WHEN DOES A TOLERANT B-REP DEFINE A SOLID? EXISTENCE, NON-EXISTENCE AND UNIQUENESS OF REALIZATIONS

GUOQING (EDDIE) ZHANG

Abstract. Boundary representations in CAD kernels carry tolerances: edge curves, the boundary curves of the adjacent faces, and vertex points do not fit together exactly, and the solid they describe is only implied. We make this precise. A *realization* of a tolerant B-rep is a PL embedding of its cell complex whose cells stay in their tolerance zones. We prove three kinds of results. (1) *Uniqueness:* if every shell has genus at least one, the carriers of the cells are contractible, and each shell has a product collar containing its zones, then all realizations are ambient isotopic, and so are the solids they bound. The collar condition reduces to a distance bound below the weak feature size of a reference surface, and for polyhedral reference surfaces we give an explicit lower bound for the weak feature size in terms of conical radii; the key lemma is machine-checked in Lean. (2) *Existence:* under computable local hypotheses there is a realization, built either by fans in the normal disks of the edges with cones at the vertices, or, for sharp edges with tight tolerances, by re-intersecting the face surfaces. (3) *Non-existence:* if two planar patches cross transversally away from the shadows of all edge and vertex zones, there is no realization. Experiments with OpenCascade show that its validity checks do not decide any of these questions. In particular, an ordinary sewing operation produces a wedge that passes both of OpenCascade's checks and has no realization, and a thick pipe that passes both checks has two realizations whose solids are not homeomorphic.

## 1. Introduction

1.1. **The problem.** A solid in a CAD kernel is stored as a boundary representation: a cell complex (faces, edges, vertices) with geometry attached to each cell. Each face carries a trimmed patch of a surface, each edge a 3D curve together with one parameter-space curve (pcurve) for every adjacent face, and each vertex a point. In floating point, and after modelling operations such as Booleans, fillets, offsets and sewing, these data do not agree: the images of the pcurves on the surfaces miss the 3D edge curve, and the edge curves miss the vertex points. Kernels in the tradition of Jackson [8] therefore attach a tolerance to every vertex, edge and face, and regard the data as consistent when the deviations lie within the tolerances. OpenCascade, for instance, requires the pcurve images to lie within the edge tolerance of the 3D curve, and the edge ends within the vertex tolerance of the vertex point [9].

What solid does such a model describe? Qi and Shapiro [11] describe the conventional answer as: the representation is valid if there exists an exact solid in the tolerance zones. Two questions follow.

(1) *Existence.* Is there an embedded surface that respects all the tolerances?
(2) *Uniqueness.* If there is more than one, do they all bound the same solid up to ambient isotopy?

Downstream algorithms (Booleans, offsets, meshing, analysis) implicitly assume both answers are yes.

1.2. **What we prove.** We call an embedding as in (1) a *realization*, and make it precise in Section 2: a PL embedding $h$ of the cell complex $K$ with $h(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$ for every cell $\sigma$, where $\mathcal{Z}(\sigma)$ is the tolerance zone of $\sigma$.

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- **Uniqueness** (Theorem 3.4). Suppose the carriers of the cells are contractible, each shell has a product collar that contains its zones, and every shell has genus at least one. Then any two realizations are ambient isotopic, and so are the solids they bound. The proof combines a carrier lemma (a cellular homotopy argument) with Waldhausen's theorem that incompressible surfaces in a product $F \times I$ are horizontal [16, Prop. 3.1].
- **The collar condition is computable** (Section 4). It holds when all zones lie within distance $d$ of a reference surface $M$ with $d < \mathrm{wfs}(M)$, the weak feature size. For a polyhedral $M$ we prove $\mathrm{wfs}(M) \geq \lambda(M) > 0$, where $\lambda$ is defined from the conical radii of $M$ (Corollary 4.3). The underlying cone lemma is machine-checked in Lean 4 with Mathlib. We also give a certified octree computation of a lower bound for $\mathrm{wfs}(M)$.
- **Existence** (Section 5). There are two constructions, each under computable hypotheses:
  - fans in the normal disks of the edges, with cones at the vertices (Theorem 5.1);
  - re-intersection of the face surfaces, for sharp edges with tight tolerances (Propositions 5.3 and 5.4).
- **Non-existence** (Theorem 6.1). If two planar patches cross transversally away from the projections ("shadows") of all edge and vertex zones, there is no realization.

1.3. **What the experiments show.** In OpenCascade 8.0.1 (Section 7):

- `BRepCheck_Analyzer` accepts models regardless of the size of the tolerances.
- `BRepAlgoAPI_Check` gives false alarms on realizable models, and misses models whose tolerances exceed the weak feature size.
- A thick pipe that passes both checks has two realizations whose solids are not homeomorphic: one ties a trefoil into the hole.
- Sewing a wedge whose faces are slightly displaced produces a model that passes both checks and, by Theorem 6.1, has no realization. The hypotheses were verified in exact rational arithmetic.

1.4. **Related work.**

- *Tolerances in practice.* Segal [13] and Jackson [8] introduced vertex and edge tolerances into polyhedral and B-rep modelling; Jackson's paper is the basis of the local-tolerance approach used by commercial kernels.
- *Rigorous validity, without tolerances.* Sakkalis, Shen and Patrikalakis [12] study interval solids: boxes covering the boundary of a known exact solid, up to homeomorphism.
- *Global tolerances.* Qi and Shapiro [10] develop $\varepsilon$-solidity, a theory with a single global tolerance that does not require an exact solid to exist [11]. Shapiro [14] studies homotopy conditions for tolerant geometric queries.
- *Constructing an exact solid.* Andersson, Stewart and Zidani [2] and Stewart and Zidani [15] construct a solid from inconsistent trimmed-patch data by perturbing the patches (Whitney extension, transfinite interpolation). They *assume* that the perturbed patches form an embedded surface: ASZ say that adjacent patches need an a posteriori check [2, pp. 819–820], and SZ make a similar assumption [15, p. 188].
- *Isotopy of approximations.* The weak feature size and its stability theory are due to Chazal and Lieutier [3]; reach is due to Federer [6].

Not yet read: Andersson, Peters and Stewart [1] on equivalence of topological form; [the 2006 paper of Andersson, Stewart and Zidani — not yet checked against the original].

*Novelty, stated cautiously.*

- Theorem 3.4 applies Waldhausen's theorem; the new part is the set-up (zones, carriers, collars) that makes it applicable to tolerant B-reps.
- Positivity of wfs for compact semialgebraic sets is known: di Rocco, Eklund and Gäfvert [4, Thm. 2.5] state it and attribute it to Fu [7] [Fu's paper — not yet checked against the

original]. We did not find the explicit bound $\lambda(M)$ in the literature we checked. That does not make it new.
- The existence results derive the embedding from checkable hypotheses rather than assuming it. As far as we have read, this is new; it remains to be checked against [1] and the 2006 ASZ paper.
- We did not find an obstruction result like Theorem 6.1 in the literature we read.

## 2. Tolerant B-reps and realizations

2.1. **Complexes.** $K$ is a finite CW complex whose underlying space $|K|$ is a disjoint union of closed connected surfaces $|K_1|, \dots, |K_n|$ (the *shells*). We write $\tau \le \sigma$ when $\tau$ is a face of $\sigma$ (including $\tau = \sigma$). For the uniqueness theorem we assume $K$ is *regular*: every closed cell $\bar\sigma$ is a ball whose boundary is a subcomplex. B-rep complexes with seam edges, closed edges or faces with holes are not regular, and must first be subdivided. The existence constructions work directly on B-rep complexes; we say so where it matters.

2.2. **Geometric data and zones.**

- Each vertex $v$ has a point $p_v$ and a tolerance $\tau_v$.
- Each edge $e$ has a $C^2$ curve $\gamma_e\colon [0,1] \to \mathbb{R}^3$ and a tolerance $\tau_e$.
- Each face $f$ has a $C^2$ surface $S_f$, a compact trimmed patch $P_f \subseteq S_f$, and a tolerance $\tau_f$. The boundary of $P_f$ consists of the images $\beta_{f,e} = S_f \circ (\text{pcurve})$ of the pcurves of the edges $e < f$.

The *deviation* of $e$ is $\mathrm{dev}_e = \max_{f>e} \max_t |\beta_{f,e}(t) - \gamma_e(t)|$. OpenCascade requires $\mathrm{dev}_e \le \tau_e$.

**Definition 2.1** (Zones)**.** We use *augmented* open zones:

- $\mathcal{Z}(v) = B(p_v, \tau_v)$;
- $\mathcal{Z}(e) = N_{\tau_e}(\gamma_e) \cup \bigcup_{v<e} \mathcal{Z}(v)$;
- $\mathcal{Z}(f) = N_{\tau_f}(P_f) \cup \bigcup_{e<f} \mathcal{Z}(e)$.

Here $N_r$ is the open $r$-neighbourhood. Augmentation makes $\mathcal{Z}$ monotone: $\tau \le \sigma$ implies $\mathcal{Z}(\tau) \subseteq \mathcal{Z}(\sigma)$.

**Definition 2.2** (Realization)**.** A *realization* of the tolerant B-rep $T = (K, \mathrm{data}, \tau)$ is a PL embedding $h\colon |K| \to \mathbb{R}^3$ with $h(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$ for every cell $\sigma$.

*Remark* 2.3.
- *Open-cell condition.* We require only $h(\operatorname{int}\sigma) \subseteq \mathcal{Z}(\sigma)$. The closed-cell condition $h(\bar\sigma) \subseteq N(\sigma)$, with unaugmented neighbourhoods, can fail for every candidate when vertex tolerances exceed edge tolerances, which is the usual ordering. With augmented zones the open-cell condition implies $h(\bar\sigma) \subseteq \mathcal{Z}(\sigma)$, since $\bar\sigma$ is the union of the open cells $\tau \le \sigma$.
- *Tameness.* Requiring $h$ to be PL rules out wild embeddings such as Alexander's horned sphere.

**Definition 2.4** (Carriers)**.** The *carrier* of a cell is $\mathcal{C}(\sigma) = \bigcup_{\tau \le \sigma} \mathcal{Z}(\tau)$. For augmented zones $\mathcal{C}(\sigma) = \mathcal{Z}(\sigma)$. A continuous map $g\colon |K| \to \mathbb{R}^3$ is *carried* if $g(\bar\sigma) \subseteq \mathcal{C}(\sigma)$ for every $\sigma$. By Remark 2.3, every realization is carried.

**Definition 2.5** (Solid)**.** Let $S_1, \dots, S_n \subset \mathbb{R}^3$ be pairwise disjoint closed surfaces. The *depth* of a point $x \notin \bigcup S_j$ is the number of $j$ for which $x$ lies in the bounded component of $\mathbb{R}^3 \setminus S_j$. The solid bounded by the $S_j$ is the closure of the set of points of odd depth.

This matches the usual B-rep convention of outer shells and cavities when the shells are consistently oriented; a consistency check of the orientation is separate.

## 3. Uniqueness for shells of positive genus

### 3.1. Hypotheses.

- **Carrier condition.** $\mathcal{C}(v)$ is path-connected for every vertex $v$, and $\pi_k(\mathcal{C}(\sigma)) = 0$ for every $k$-cell $\sigma$ with $k \geq 1$. Contractible carriers suffice.
- **Collar condition.** For each shell $K_i$ there are a compact PL submanifold $U_i \subset \mathbb{R}^3$ and a PL homeomorphism $U_i \cong M_i \times [-1, 1]$ with these properties:
  - $M_i \times \{0\}$ is the image of a carried PL embedding $\varphi_i \colon |K_i| \to \mathbb{R}^3$ (a *reference surface*);
  - all zones of the cells of $K_i$ lie in $\operatorname{int} U_i$;
  - the $U_i$ are pairwise disjoint.
- **Genus.** Each $M_i$ has genus at least one.

### 3.2. Lemmas.

**Lemma 3.1** (Carrier lemma)**.** *Under the carrier condition, any two carried maps $f, g \colon |K| \to \mathbb{R}^3$ are homotopic by a homotopy $H$ with $H(\bar{\sigma} \times I) \subseteq \mathcal{C}(\sigma)$ for every $\sigma$.*

*Proof.* Define $H$ on $|K| \times \{0, 1\} \cup |K^{(k)}| \times I$ by induction on $k$, keeping $H(\bar{\sigma} \times I) \subseteq \mathcal{C}(\sigma)$, with $H(\cdot, 0) = f$ and $H(\cdot, 1) = g$.

- $k = 0$: for a vertex $v$, the points $f(v), g(v)$ lie in the path-connected set $\mathcal{C}(v)$; join them by a path in $\mathcal{C}(v)$.
- $k \geq 1$: let $\sigma$ be a $k$-cell. $\bar{\sigma} \times I$ is a $(k+1)$-ball whose boundary $\bar{\sigma} \times \{0, 1\} \cup \partial\bar{\sigma} \times I$ is a $k$-sphere, and $H$ is already defined on it:
  - on $\bar{\sigma} \times \{0, 1\}$ it is $f$ or $g$, with image in $\mathcal{C}(\sigma)$;
  - on $\partial\bar{\sigma} \times I$ it is defined by induction, with image in $\bigcup_{\tau < \sigma} \mathcal{C}(\tau) \subseteq \mathcal{C}(\sigma)$.

  Since $\pi_k(\mathcal{C}(\sigma)) = 0$, $H$ extends over $\bar{\sigma} \times I$ with image in $\mathcal{C}(\sigma)$.

The pieces agree on common boundaries, and a finite union of closed sets glued continuously gives a continuous $H$. □

**Lemma 3.2** (Separation)**.** *Assume the carrier and collar conditions, and let $h$ be a realization. Then $G = h(|K_i|)$ is a closed surface in $\operatorname{int} U_i$, and the inclusion $G \hookrightarrow U_i$ is a homotopy equivalence.*

*Proof.*

- By Lemma 3.1 on $K_i$, $h$ and $\varphi_i$ are homotopic through maps with image in $\bigcup \mathcal{C}(\sigma) \subseteq \operatorname{int} U_i$, so the homotopy takes place in $U_i$.
- $\varphi_i \colon |K_i| \to U_i$ is a homeomorphism onto $M_i \times \{0\}$ followed by inclusion, which is a homotopy equivalence. Hence so is $h \colon |K_i| \to U_i$.
- Since $h$ is a homeomorphism onto $G$, the inclusion $G \hookrightarrow U_i$, which equals $h \circ h^{-1}$, is a homotopy equivalence.

□

**Lemma 3.3** (Horizontality)**.** *Under the hypotheses of Lemma 3.2, if $M_i$ is not a sphere, there is an ambient isotopy of $\mathbb{R}^3$, supported in $U_i$ and fixed on $\partial U_i$, that carries $G$ to $M_i \times \{0\}$.*

*Proof.* (a) *$G$ is incompressible.*

- By Lemma 3.2, $\pi_1(G) \to \pi_1(U_i)$ is an isomorphism.
- The boundary of a compressing disk would be a curve that is essential in $G$ and null-homotopic in $U_i$, which this rules out. That is incompressibility in the sense of Waldhausen [16, p. 58].
- Case (b) of that definition concerns spheres and does not arise. We do not use the converse direction, which needs the loop theorem.
- $G$ is a closed orientable surface in an orientable 3-manifold, hence two-sided, and it is disjoint from $\partial U_i$; so it satisfies Waldhausen's conventions [16, §0].

(b) *$G$ can be made horizontal.*

- Apply Waldhausen's Proposition 3.1 [16, pp. 65–66] to $U_i = M_i \times I$ ($M_i$ orientable, not a sphere) and the incompressible surface $G$. Since $\partial G = \emptyset$, the condition "$\partial G \subseteq F \times 1$" holds vacuously.
- It gives an isotopy, constant on $\partial U_i$, from $G$ to a surface $G'$ whose projection $p\colon G' \to M_i$ is a homeomorphism onto its image.
- $p(G')$ is open (invariance of domain) and closed in the connected surface $M_i$, so $p(G') = M_i$. Hence $G'$ is the graph of a function $s\colon M_i \to (-1, 1)$.

(c) *Flatten.* The PL homeomorphisms of the fibres $\{x\} \times [-1, 1]$ that fix both ends and send $s(x)$ to $(1 - t)s(x)$, $t \in [0, 1]$, give an ambient isotopy of $U_i$, fixed on $\partial U_i$, from $G'$ to $M_i \times \{0\}$. This isotopy is topological. Alternatively, the region between $G'$ and $M_i \times \{0\}$ is a product region, so the two surfaces are parallel [16, p. 58].

(d) Both isotopies are fixed on $\partial U_i$ and extend by the identity to $\mathbb{R}^3$.

□

### 3.3. **The theorem.**

**Theorem 3.4** (Uniqueness)**.** *Assume the carrier condition, the collar condition, and that every shell has genus at least one. Then:*

(1) *the image of any realization is ambient isotopic to* $\bigcup_i M_i \times \{0\}$*;*
(2) *the images of any two realizations are ambient isotopic;*
(3) *the solids they bound (Definition 2.5) are ambient isotopic.*

*Proof.*

- Apply Lemma 3.3 shell by shell. The supports $U_i$ are pairwise disjoint, so the isotopies can be run simultaneously. Their end map is a homeomorphism $\Phi$ of $\mathbb{R}^3$ with $\Phi(G_i) = M_i$ for every $i$.
- By the Jordan–Brouwer separation theorem each closed surface separates $\mathbb{R}^3$ into two components. $\Phi$ maps the bounded component of $\mathbb{R}^3 \setminus G_j$ onto the bounded component of $\mathbb{R}^3 \setminus M_j$. So depth is preserved, and $\Phi$ maps the solid bounded by the $G_j$ onto the solid bounded by the $M_j$.
- For two realizations with maps $\Phi$ and $\Phi'$, use $\Phi'^{-1} \circ \Phi$.

□

*Remark* 3.5.

- *What is not covered.* Spheres (genus 0) are excluded, and the isotopy is of images, not of embeddings.
- *The collar condition cannot be dropped.* The thick pipe of Section 7 passes both OpenCascade checks and has two realizations whose solids are not even homeomorphic.

## 4. The collar condition is computable

### 4.1. **Weak feature size.**

Let $M \subset \mathbb{R}^3$ be compact, $d(p) = \operatorname{dist}(p, M)$, and let $\Gamma(p)$ be the set of nearest points of $p$ in $M$. A point $p \notin M$ is *critical* if $0 \in \operatorname{conv}\{p - x : x \in \Gamma(p)\}$. The *weak feature size* $\operatorname{wfs}(M)$ is the infimum of the positive critical values of $d$ [3]. Unlike the reach, it stays positive at sharp edges and small fillets. A surface with a concave edge has reach 0, and a fillet of radius $r$ caps the reach at $r$.

**Lemma 4.1** (Collar)**.** *Let $M$ be a closed PL surface in $\mathbb{R}^3$, and let $0 < r < \operatorname{wfs}(M)$.*

(a) *The closed $r$-neighbourhood $\bar{N}_r(M)$ is PL-homeomorphic to $M \times [-1, 1]$, with $M$ going to $M \times \{0\}$.*
(b) *Suppose every zone of every cell of a shell lies in $N_d(M)$ for some $d < r$, $M$ is the image of a carried PL embedding of that shell, and the neighbourhoods $\bar{N}_r(M_i)$ of different shells are pairwise disjoint. Then the collar condition of Section 3 holds with $U_i = \bar{N}_r(M_i)$.*

*Proof sketch.*
- *Small radii.* For small $\delta > 0$, $\bar{N}_\delta(M)$ is a regular neighbourhood of $M$ relative to $M$, by Durfee's theorem on neighbourhoods of semialgebraic sets [5, Props. 3.5, 3.7]. A regular neighbourhood of a closed orientable surface in $\mathbb{R}^3$ is a product.
- *Larger radii.* The region $\{\delta \le d \le r\}$ contains no critical point, so it is a product of a level set with $[\delta, r]$, by the isotopy lemma of Chazal and Lieutier [3, Prop. 3.4].
- *Gluing.* Gluing the two products gives (a). Part (b) follows directly.

[Draft note: The application of [3, Prop. 3.4], which is stated for distance to the complement of a bounded open set, to $d = \operatorname{dist}(\cdot, M)$ is spelled out in the repository notes (review of Lemma 1); it should be written out here. The PL (rather than topological) form of (a) also needs a sentence.] □

4.2. **An explicit lower bound for polyhedral surfaces.** For $a \in M$, the *conical radius* $\rho(a)$ is the supremum of the $r > 0$ such that $M \cap B(a, r) = C \cap B(a, r)$ for some closed cone $C$ with apex $a$.

**Lemma 4.2** (Cone lemma)**.** *Let $a \in M$, $r < \rho(a)$, and $p \notin M$ with $|p - a| < r/2$. Then every nearest point $x \in \Gamma(p)$ satisfies $\langle p - x,\, p - a\rangle = |p - x|^2$. In particular $p$ is not critical.*

*Proof.*
- *Nearest points lie in the ball.* $d(p) \le |p - a| < r/2$, so every $x \in \Gamma(p)$ has $|x - a| < r$.
- *The ray through $x$ stays in $M$.* The points $y_s = a + s(x - a)$, for $s$ near 1, lie in the cone and in $B(a, r)$, hence in $M$. So $s \mapsto |p - y_s|^2$ has a local minimum at $s = 1$, which gives $\langle p - x, x - a\rangle = 0$.
- *Conclusion.* $\langle p - x, p - a\rangle = |p - x|^2 + \langle p - x, x - a\rangle = d(p)^2 > 0$ for all $x \in \Gamma(p)$. So 0 is not in the convex hull of the vectors $p - x$.

□

**Corollary 4.3.** *Let $\lambda(M) = \inf_{x \in M} \sup_{a \in M} \big(\rho(a)/2 - |x - a|\big)$. Then $d$ has no critical point $p$ with $0 < d(p) < \lambda(M)$, so $\operatorname{wfs}(M) \ge \lambda(M)$. If $M$ is a finite union of triangles, $\lambda(M) > 0$.*

*Proof.*
- *The bound.* Given $p$ with $d(p) < \lambda$ and $x \in \Gamma(p)$, pick $a$ with $\rho(a)/2 - |x - a| > d(p)$. Then $|p - a| < \rho(a)/2$, and Lemma 4.2 applies.
- *Positivity.* For a polyhedron, $\rho(a) > 0$ for every $a$. The function $g(x) = \sup_a(\rho(a)/2 - |x - a|)$ is 1-Lipschitz and finite, since $\rho \le \operatorname{diam} M$, and $g(x) \ge \rho(x)/2 > 0$. $M$ is compact, so $\min g > 0$.

□

Lemma 4.2 and the pointwise form of Corollary 4.3 are machine-checked in Lean 4 with Mathlib, for subsets of an arbitrary real inner product space (Appendix A).

4.3. **A certified lower bound for** wfs**.** For a triangulated reference surface we certify "no critical value in $(0, L)$" with a best-first octree:

- *Boxes near $M$* are certified by Lemma 4.2, using a lower bound for $\rho(a)$ at a candidate point $a$. The bound comes from the distance from $a$ to the nearest feature (planar region, boundary segment, vertex) that does not contain $a$.
- *Boxes farther away* are certified by a direction test: a small linear program finds a unit vector $u$ with $\langle u, p - x\rangle > 0$ for all candidate nearest points $x$ of all points $p$ of the box. The candidates are the projections of the box centre onto the relevant features.
- *Stopping rule.* The search stops at the first box that is too small to split further and cannot be certified. Its distance from $M$ is the certified bound $L$.

The computation is in double precision, not interval arithmetic. Table 1 lists the results.

## 5. Existence

We give two constructions. Both turn computable hypotheses on the data into a realization. Neither assumes that some perturbed surface embeds.

| Part | exact wfs | certified lower bound |
|---|---|---|
| cube, side 10 | 5 | 4.530 |
| plate $4 \times 4 \times 1$ | 0.5 | 0.436 |
| plate $4 \times 4 \times 0.3$ | 0.15 | 0.137 |
| L-shape (wall 2) | 1 | 0.894 |
| L-shape, inner fillet 0.2 | 1 | 0.894 |
| block with slot of width 0.2 | 0.1 | 0.086 |
| cylinder $R = 3$, $H = 10$ | 3 | 2.512 |

TABLE 1. Certified lower bounds for wfs of the triangulated parts. The inner fillet caps the reach at 0.2 but leaves the certified wfs bound unchanged.

### 5.1. **Fans and cones.**

*Auxiliary sets.*

- *Normal disks.* For an edge $e$, choose a $C^1$ radius function $\rho_e(t)$ and parameters $0 < a_e < b_e < 1$. $D_e(t)$ is the closed disk of radius $\rho_e(t)$ in the normal plane $\Pi(t)$ of $\gamma_e$ at $\gamma_e(t)$, and $T_e = \bigcup_{t\in[a_e,b_e]} D_e(t)$.
- *Vertex regions.* For a vertex $v$, choose $r_v$. The vertex region $W_v$ is the union of the ball $B(p_v, r_v)$ and, for each edge $e$ at $v$, the disks $D_e(t)$ for $t$ between the end at $v$ and $a_e$ (or $b_e$).

*Hypotheses.* The full statements, with the constants, are in the repository (`notes/lemma0_v02.md`, `notes/lemma0_v03.md`). In summary:

**E1:** $\operatorname{dev}^*_e(t) < \rho_e(t) < \min(\tau_e, \operatorname{reach}\gamma_e)$, where $\operatorname{dev}^*_e(t)$ is the deviation measured in the normal plane. $\rho_e$ is constant on the stubs near the vertices.

**E2:** Each adjacent face crosses the wall of $T_e$ exactly once in every normal plane. Sufficient conditions are given in terms of the angle between $\gamma'_e$ and the face, the curvature of the face, and a no-return distance.

**E3′:** In every normal plane, the exit points $k_{f,e}(t)$, $k_{g,e}(t)$ of the two face sections are distinct. (E3 is an explicit sufficient condition: the angle between the sections exceeds $\varepsilon_f + \varepsilon_g$, where $\varepsilon = \arcsin\big((\operatorname{dev}^* + k\ell^2/2)/\rho\big)$.)

**V0, V1:** The edge ends and patch corners at $v$ lie in $B(p_v, r_v)$. $W_v$ is strictly star-shaped from $p_v$ and lies in $\mathcal{Z}(v)$. The construction meets $\partial W_v$ in one simple closed curve $\lambda_v$, in the cyclic order of the link of $v$; this part is checked computationally.

**S1, P0:** Separation of the $T_e$, the $W_v$ and the patches, and each patch with its collar removed is a disk. These are checked computationally.

**Theorem 5.1** (Existence by fans and cones)**.** *Under E1–E3′, V0, V1, S1 and P0, the tolerant B-rep has a realization.*

*Construction and proof outline.* (1) *Vertices and edges.* Put $h(v) = p_v$. On $[a_e, b_e]$, let $h(e)$ follow $\gamma_e$.

(2) *Fans.* In each disk $D_e(t)$, the radius from $\gamma_e(t)$ to $k_{f,e}(t)$ is assigned to $f$. These radii sweep out the fan strip $F_{f,e}$.

(3) *Faces away from the vertices.* Outside the vertex regions, the face is $Q_f \cup \bigcup_e F_{f,e}$, where $Q_f$ is $P_f$ with the tubes and vertex regions removed.

(4) *Vertex regions.* Inside $W_v$, take the cone from $p_v$ over $\lambda_v$.

(5) *Injectivity and zones.* The disks $D_e(t)$ are disjoint (E1). Within one disk there are two radii meeting only at the centre (E3′). Nothing else enters the tubes or vertex regions (S1,

P0). The cone is injective by star-shapedness. Each piece lies in the zone of its cell with a positive margin.

(6) *PL step.* The resulting map is bi-Lipschitz: every crease has a positive opening angle, and at points where three pieces meet the tangent cone is an embedded disk. A fine secant (vertex-interpolating) approximation on a shape-regular triangulation is then injective and stays in the zones.

Two independent reviews of v0.2 and three of v0.3 found no fatal error. □

**Example 5.2.** On a cube whose faces are pulled apart by gaps, the construction was carried out exactly in rational arithmetic: 216 triangles, with zone membership and embeddedness checked exactly (`compute/lemma0_box.py`).

### 5.2. **Re-intersection.**

**Proposition 5.3** (Planar, degree three)**.** *Suppose every face is planar, $P_f \subseteq \Pi_f$, and every vertex lies on exactly three faces whose planes are independent. Let $c_v$ be the common point of the three planes, $\ell_e = [c_v, c_w]$ for an edge $e = vw$, and $\tilde{P}_f$ the polygon on the $c$'s of the vertices of $f$, in the order of $\partial f$. Assume:*

- *$c_v \in \mathcal{Z}(v)$, $\ell_e \subseteq \mathcal{Z}(e)$ and $\tilde{P}_f \subseteq \mathcal{Z}(f)$;*
- *the $\tilde{P}_f$ are simple polygons that meet exactly in their common edges and vertices.*

*Then the cellular map onto $\bigcup \tilde{P}_f$ is a realization.*

*Proof.* It is a PL homeomorphism onto $\bigcup \tilde{P}_f$, and each cell lands in its zone. □

**Proposition 5.4** (Curved, transversal)**.** *Suppose every face lies on a $C^2$ surface $S_f$, and there is a* re-trimmed model*: patches $\tilde{P}_f \subseteq S_f$, curves $\tilde{\gamma}_e$ and points $c_v$ bounding a solid, whose cell structure is isomorphic to $K$ (as a B-rep complex, respecting seams and closed edges). Assume:*

- *the closed cells lie in their zones;*
- *the cells are $C^2$ up to the boundary, the curves are regular, and there are no cusp corners;*
- *the surfaces meet at a positive angle along every non-seam edge;*
- *at every vertex the tangent cone is an embedded PL disk.*

*Then a realization exists.*

*Proof.*

- Take $|K|$ to be the re-trimmed surface with its cells, and $h_0$ the identity.
- By compactness, $h_0(\bar{\sigma})$ has a positive margin $m$ inside the open zone $\mathcal{Z}(\sigma)$.
- $h_0$ is bi-Lipschitz: within a cell by regularity, across an edge by the positive crease angle, and at a vertex by the tangent-cone condition.
- A secant approximation on a fine shape-regular curved triangulation is injective once its Lipschitz distance to $h_0$ is below the bi-Lipschitz constant, and it stays in the zones once its distance is below $m$.

□

*Remark* 5.5 (Checking the hypotheses).

- The re-trimmed model can be computed by extending the face surfaces and rebuilding the solid (OpenCascade's `BOPAlgo_MakerVolume`). The hypotheses are then checked on samples (`compute/reintersect_curved.py`).
- For the planar case, the hypotheses of Proposition 5.3 for the sewn box were also re-checked in exact rational arithmetic, including embeddedness.
- Results are in Section 7.

## 6. An obstruction

### 6.1. **Setting.**

- $f \neq g$ are faces whose closed cells are disks bounded by single wires.

- Their data are planar: $P_f \subseteq \Pi_f$ and $P_g \subseteq \Pi_g$, with the planes not parallel.
- $\lambda = \Pi_f \cap \Pi_g$, and $\alpha \in (0, \pi/2]$ is the angle between the planes.
- $n_f, n_g$ are unit normals, and $m_f$ is a unit vector in $\Pi_f$ orthogonal to $\lambda$ (likewise $m_g$).
- $\pi_f$ is the orthogonal projection onto $\Pi_f$, and $U_f = \bigcup_{e<f} \mathcal{Z}(e)$, which contains the zones of the vertices of $f$.
- $\ell_f$ is the boundary loop of $f$: the arcs $\beta_{f,e}$ in the cyclic order of $\partial f$, with consecutive arcs joined by the segment between their end points.

**Theorem 6.1** (Obstruction)**.** *Suppose there are $X \in \lambda$ and $\rho > 0$ such that, with $J_f = X + [-\rho, \rho]m_f$ and $J_g = X + [-\rho, \rho]m_g$:*

**O0:** *$\beta_{f,e} \subseteq \mathcal{Z}(e)$ for every edge $e$ of $f$, and at every vertex $v$ of $f$ the end points of the two arcs of $f$ at $v$ lie in $\mathcal{Z}(v)$; likewise for $g$.*

**O1:** *$J_f \cap \ell_f = \emptyset$, and the winding number of $\ell_f$ in $\Pi_f$ about the points of $J_f$ is odd; likewise for $g$.*

**O2:** *$J_f \cap \operatorname{cl} \pi_f(U_f) = \emptyset$ and $J_g \cap \operatorname{cl} \pi_g(U_g) = \emptyset$.*

**O3:** *$\mathcal{Z}(e)$ is simply connected for every edge $e$ of $f$ or $g$.*

**O4:** *$\rho \sin\alpha > \tau_f \cos\alpha + \tau_g$ and $\rho \sin\alpha > \tau_g \cos\alpha + \tau_f$.*

*Then the tolerant B-rep has no realization.*

*Proof.* Suppose $h$ is a realization, and let $\varphi = \pi_f \circ h|_{\bar{f}}$.

*Step 1: over $J_f$, $h(\bar{f})$ lies in the slab.*

- Let $y \in J_f$. We have $h(\bar{f}) \subseteq \mathcal{Z}(f) = N_{\tau_f}(P_f) \cup U_f$, and by O2 no point of $U_f$ lies over $y$.
- Since $P_f \subseteq \Pi_f$, $\pi_f^{-1}(y) \cap N_{\tau_f}(P_f) \subseteq \sigma_y := y + (-\tau_f, \tau_f)n_f$.
- So $h(\bar{f}) \cap \pi_f^{-1}(y) = h(\bar{f}) \cap \sigma_y$. These points are images of interior points of $f$, because $h(\partial f) \subseteq U_f$.

*Step 2: degree one.*

- Parametrize $\ell_f$ over $\partial f$: a small arc $A_v$ around each vertex goes to the joining segment, and the rest of each edge goes to $\beta_{f,e}$. Shrink $A_v$ until $h(A_v) \subseteq \mathcal{Z}(v)$.
- Build a homotopy in $U_f$ from $h|_{\partial f}$ to this parametrization:
  - on $A_v$, the straight-line homotopy inside the convex ball $\mathcal{Z}(v)$;
  - on the rest of an edge $e$, the boundary of the square is mapped into $\mathcal{Z}(e)$ (by O0, monotonicity, and $h(\bar{e}) \subseteq \mathcal{Z}(e)$), so it extends over the square by O3.
- By O2 the projected homotopy avoids every $y \in J_f$. So $\varphi|_{\partial f}$ and $\ell_f$ have the same winding number about $y$, which is odd by O1. Hence the mod-2 degree of $\varphi$ at $y$ is 1.
- Fix a triangulation of $\bar{f}$ on which $h$ is linear. For $y \notin \varphi(\text{1-skeleton})$, which excludes only finitely many points of $J_f$ once the slice in Step 3 is generic, $\#(h(\bar{f}) \cap \sigma_y)$ is odd.

*Step 3: a crossing arc.*

- Let $\Sigma$ be the plane through $X$ orthogonal to $\lambda$; it contains $n_f, m_f, n_g, m_g$.
- Move $X$ along $\lambda$ and shrink $\rho$ slightly to generic values; O1, O2 and O4 are open conditions. Genericity means: $\Sigma$ contains no vertex of the triangulations and meets their edges transversally, and the lines $\langle x - X, m_f \rangle = \pm\rho$, $\langle x - X, m_g \rangle = \pm\rho$ contain no vertex.
- In $\Sigma$, the rectangle $R_f = J_f + [-\tau_f, \tau_f]n_f$ meets $h(\bar{f})$ in a compact 1-manifold $M_f$.
- $M_f$ misses $h(\partial f)$, since $\pi_f(R_f) = J_f$ and O2 holds. It misses the long sides of $R_f$, by Step 1. So its boundary lies on the two short sides.
- A generic fibre $\sigma_y$ meets $M_f$ an odd number of times (Step 2). Circle components, and arcs with both ends on one short side, meet it an even number of times.
- Hence some arc $a_f \subseteq M_f$ joins the two short sides of $R_f$. Likewise there is an arc $a_g$ for $g$.

*Step 4: the arcs meet.*

- In $\Sigma$, use the coordinates $u = \langle x - X, n_f \rangle$ and $w = \langle x - X, n_g \rangle$.

- On the short sides of $R_f$, $|w| \geq \rho \sin\alpha - \tau_f \cos\alpha > \tau_g$ by O4, with opposite signs on the two sides. So $a_f$ contains a sub-arc in the parallelogram $Q = \{|u| \leq \tau_f, |w| \leq \tau_g\}$ joining $w = -\tau_g$ to $w = \tau_g$. Symmetrically, $a_g$ contains a sub-arc in $Q$ joining $u = -\tau_f$ to $u = \tau_f$.
- By O4, $Q$ lies in both rectangles.
- Two paths in a square joining the two pairs of opposite sides intersect [Maehara 1984, cited for this square lemma — not yet checked against the original]. So $h(\bar{f}) \cap h(\bar{g}) \neq \emptyset$ at a point that is the image of interior points of both $f$ and $g$ (Step 1). This contradicts injectivity.

□

*Remark* 6.2.
- *Which hypotheses bind.* Face tolerances are tiny (OpenCascade's default is $10^{-7}$), so O4 is mild. The binding conditions are O1 and O2: the patches must cross away from the shadows of the edge and vertex zones.
- *Curved patches.* The proof uses planarity in Steps 1 and 2. A version for $C^2$ patches with $\tau_f$ below their reach should follow with tubular coordinates; it is not written.
- *Review.* The proof was reviewed twice independently; the second review found the proof correct and one gap in the statement, now fixed (the single-wire assumption).

## 7. Experiments with OpenCascade

### 7.1. Set-up.

- *Software.* All experiments use OpenCascade 8.0.1 through `cadquery-ocp`. Scripts and results are in the repository [17].
- *The two checks.* "BRepCheck" is `BRepCheck_Analyzer(shape).IsValid()`. "BOP check" is `BRepAlgoAPI_Check(shape, true, true)`, which also tests geometry and self-interference.
- *Sewn models* imitate imported data. Every face of an exact part is moved by a random vector of length between $g/2$ and $g$, and the faces are sewn back with `BRepBuilderAPI_Sewing` at tolerance $4g$. Sewing sets each edge tolerance to 1.05 times the measured deviation.
- *Scope.* These are a few parts in one kernel version, not a survey of industrial models.

### 7.2. The checks do not decide existence or uniqueness.

Table 2 summarises.

| Model | BRepCheck | BOP check | What is true |
|---|---|---|---|
| twisted triangular prism, two faces with linked boundaries ($\tau = 0.081$) | valid | self-intersection | no realization (linking number $-1$) |
| thick pipe, tolerance $1.3\times$ hole radius | valid | valid | two realizations with non-homeomorphic solids (knot determinant 3 vs. 1) |
| sewn wedges, 30° and 10°, $g = 0.01$ | valid | valid | no realization (Theorem 6.1) |
| sewn plate, thickness 0.05, $g = 0.01$ | valid | invalid | realizable (Proposition 5.3) |
| filleted L-shape, $\tau = 0.1 <$ wfs $= 1$ | valid | self-intersection | collar condition certified |
| long cylinder, tolerance above wfs | valid | valid | collar condition fails for the nominal surface |

Table 2. OpenCascade's checks against what the theory shows.

*Twisted prism.*

- Two non-adjacent triangular faces have boundary loops with linking number $-1$ (by crossing count, and by the Gauss integral: $-1.000000009$). The loops are farther apart than $2\tau$.
- In any realization, the image boundary loops stay in disjoint tubular neighbourhoods and are homotopic to the cores, so they are still linked. Hence each meets the other's face, contradicting embeddedness.

*Knotted pipe.*

- One realization keeps the nominal geometry. The other replaces the inner face by a thin tube, of radius 0.005, around a centreline that follows a long trefoil.
- The script checks the zones (maximum distance $0.983 \leq 1.3$), embeddedness ($\varepsilon\kappa_{\max} = 0.31$; non-local distance $0.036 > 2\varepsilon$) and the knot (determinant 3).
- The second solid is the exterior of a knotted arc in a ball, whose fundamental group is non-abelian. The nominal solid is a solid torus. So the two solids are not homeomorphic.
- This shows that the collar hypothesis of Theorem 3.4 cannot be dropped. Here $\mathrm{wfs} \leq 1 < 1.3$.

*Sewn wedges.*

- At the knife edge of the sewn wedges, the two planar patches cross along a line that lies outside the edge tube, by 49% (30°) and about 3× (10°) of $\tau_e$.
- The hypotheses of Theorem 6.1 hold. For the 30° wedge the largest feasible $\rho$ is $1.6 \cdot 10^{-3}$, while O4 needs only $3.7 \cdot 10^{-7}$; for the 10° wedge it is $1.6 \cdot 10^{-2}$.
- An independent check in exact rational arithmetic confirmed all of O0–O4 for both wedges.
- On the sewn box and plate and on the exact box and wedge, the checker finds no crossing witness, consistent with their being realizable.

| Model | fans (Thm. 5.1) | re-intersection | margins of re-intersection (vertex / edge / face) |
|---|---|---|---|
| exact box, cylinder, pipe, slot, filleted box, 10° wedge | pass | pass where applicable | — |
| L-shape, all edges filleted 0.5 | pass with variable radius* | not applicable | — |
| L-shape fused, inner fillet 0.2 | fails (9 cusp vertices); passes after merging coplanar faces | not applicable | — |
| sewn box $g = 0.01$ | fails at vertices | pass | 45% / 13% / 5.9% |
| sewn plate $g = 0.01$ | fails at vertices | pass | 45% / 13% / 5.9% |
| sewn cylinder $g = 0.01$ | fails at vertices | pass (curved) | 55% / 35% / 5.0% |
| sewn cylinder $g = 0.05$ | fails at vertices | pass (curved) | 55% / 11% / 8.9% |
| sewn pipe $g = 0.01$ | fails at vertices | pass (curved) | 53% / 16% / 5.1% |
| sewn wedges | fails at vertices | fails (edge outside tube) | no realization (Thm. 6.1) |
| sewn filleted box $g = 0.001$ | fails at vertices | not applicable (tangent faces) | open |

TABLE 3. Constructions on exact and sewn models (sampled checks, double precision). *Provisional: the checker does not yet implement the constant radius near the vertices. Face margins are an independent dense re-check with exact projections.

### 7.3. **Which construction applies.**

- *Why the fans fail on sewn data.* The fan construction fails at the vertices of sewn models. Sewing sets $\tau_e = 1.05\,\mathrm{dev}_e$, and the tube radius must exceed the deviation; but at a corner of a box three tubes of radius greater than dev do not fit disjointly into a vertex ball of radius about $\tau_e$.
- *Re-intersection.* Re-intersection avoids this, but needs transversal faces.
- *Sewn models with tangent edges remain open.* On the sewn filleted box, OpenCascade sets each vertex tolerance equal to the largest incident edge tolerance, the patch corners lie up to $0.84\,\tau_v$ from the vertex point, and none of our vertex constructions fits.

### 7.4. **Other observations.**

- *Thickness heuristics.* Half the minimum distance between non-adjacent faces is a common estimate of the safe tolerance. It overestimates the cylinder (5 against wfs = 3), and underestimates an L-shape with a 0.05 inner fillet by a factor of about 97.
- *Offsets.* In its default mode, `BRepOffsetAPI_MakeOffsetShape` returned solids with wrong volume (errors from 44% to over a thousandfold) for a filleted box at offsets within the wfs-safe range, and BRepCheck accepted them. The documentation asks only for a "small enough" offset.
- *Public documentation.* We did not find a check that tolerances leave the topology unchanged in the public documentation of OpenCascade, or in third-party documentation of ACIS and Parasolid. That says only that the documentation does not mention one.

## 8. Limitations and open problems

(1) **Spheres.** Theorem 3.4 excludes shells of genus 0. Waldhausen's horizontality argument does not apply to them. The PL Schoenflies theorem should give the sphere case under the same hypotheses; this is not written.
(2) **Embeddings.** We prove that the images of realizations are ambient isotopic, not that the embeddings are isotopic as maps. The two differ by a self-homeomorphism of the surface.
(3) **Tight vertices.** We do not know whether sewn models with tangent edges and OpenCascade's vertex tolerances (equal to the largest incident edge tolerance) have realizations. Two directions are open: a vertex construction that uses the edge tubes as well as the ball, or a vertex version of the obstruction.
(4) **Mixed vertices.** Vertices where tangent and sharp edges meet need a construction combining fans and re-intersection.
(5) **Curved obstruction.** Theorem 6.1 is stated for planar patches.
(6) **Rigour of the computations.**
    - The wfs bounds and the hypothesis checks use double precision and sampling. Exceptions: the gapped-cube construction, and the exact re-verification of the sewn wedges and the sewn box.
    - Interval arithmetic and a compiled implementation are needed for industrial models.
(7) **Formalization.** The carrier lemma in general (contractible carriers), Waldhausen's theorem and the existence constructions are not formalized.

[Draft note: Before submission: check against the originals Moise (Geometric Topology in Dimensions 2 and 3, ch. 17, 35), Hempel (3-Manifolds, ch. 6, 10), Fu (1985), Maehara (1984), Andersson–Peters–Stewart (2000) and Andersson–Stewart–Zidani (2006). Revise the novelty statements in Section 1 accordingly.]

## Appendix A. Formal verification in Lean

The following are proved in Lean 4 with Mathlib (repository directory `lean/`). The build has no `sorry`, and the results depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

| Lean statement | Content |
|---|---|
| `inner_eq_zero_of_conical` | at a nearest point $x$, $\langle p-x, x-a\rangle = 0$ (Lemma 4.2) |
| `inner_sub_eq_norm_sq_of_conical` | $\langle p-x, p-a\rangle = \lvert p-x\rvert^2$ |
| `not_isCritical_of_conical` | $p$ is not critical (Lemma 4.2) |
| `not_isCritical_of_conicalCover` | pointwise form of Corollary 4.3 |
| `carrier_mono`, `carrier_eq_zone_of_mono` | carriers are monotone, and equal the zones when the zones are augmented |
| `carried_of_realization`, `image_closedCell_subset_carrier` | realizations are carried |
| `carried_straightLine`, `continuous_straightLine` | Lemma 3.1 when the carriers are convex |

The statements are for an arbitrary real inner product space. Criticality is formalized as $0 \in \operatorname{conv}\{p - x : x \text{ nearest}\}$.

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